Tutorials (SC/CE/CZ1006)
Tut 6

Important concepts & methodology in this week

- *Major ways of data transfers*
  - *Polling*
  - *Interrupts*
  - *DMA (direct memory access)*
- *Modes & factors affecting DMA efficiency*

1)

1/ (10+90) μs = 10,000 ISRs serviced per second

Each character transfer involved 10 bits on UART (S-7D-P-STP).

10 bits * 10000 = 100 Kbits per second

2)

(a)

Polled every 100ms, each poll takes 10ms ⇨ 90ms(idle), 10ms(active).

⇨ Average current = 0.9*0.1mA + 0.1*50mA = 5.09mA

(b) Interrupt mechanism. ⇨ Interrupted 400 times over 24-hour

⇨ Active time = 400*20ms = 8000ms

⇨ Average current consumed over 24-hour = [8*50mA + ((24*60*60)-8)*0.1mA] / [24*60*60] = 0.1046mA

(c) Polling

i. Average current consumed 5.09mA

ii. Battery life = 2000mAh/5.09mA = 393 hours = 16.4 <u>days</u>

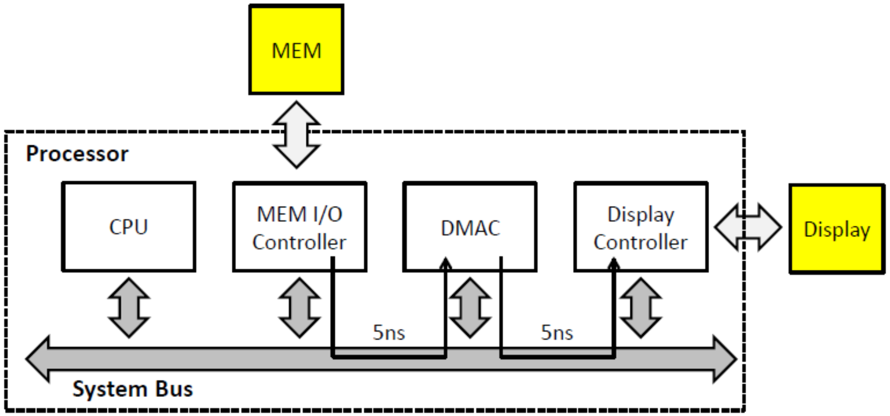
Interrupt

i. Average current consumed 0.1046 mA

ii. Battery life = 2000mAh/0.1046mA = 19120 hours = 797 days = 2.2 <u>years</u>

3)

- CPU competes with the DMAC (DMA Controller) where **system bus utilization** is concerned. Higher CPU utilization rate (of the system bus) means lesser chance for DMAC to use the same bus, hence DMA data transfer rate will be affected.
- The competition for resources is also affected by **the priority of DMAC** with respect to the CPU. Depending on processor design, priority of CPU with respect to DMAC can be fixed or configurable. The one with higher priority will have higher chance of hitting their maximum possible transfer rate.
- Maximum Data transfer rate **is limited by the slowest device** in the whole data path consisting of the source device, DMAC and destination device. If there is a memory or I/O device that is slow, it'll affect the overall DMA transfer rate.
- • All things equal, the **mode of transfer** used by the DMAC (burst, cycle stealing or transparent) will affect the DMAC transfer rate.



4) (a) burst mode

- Each transfer of the system bus will transfer 3 bytes of video data (RGB) and take **2*5ns.**
- Number of transfers in 1/30 seconds = 1920*1080 = 2,073,600 transfers.
- Total time taken
- = Bus control request + actual transfer + Bus release
- = 100ns + 2073600*2*5ns + 100ns = 0.0207362 sec < 1/30sec
- ⇨ DMAC is able to transfer each frame of video before the start of the next frame request.

(b) cycle-stealing

- Time taken for one transfer
- = Bus control request + 1 transfer + Bus release + 5*CPU Clock cycles wait
- = 100ns + 2*5ns + 100ns + 5*(1/400Mhz) = 222.5ns
- Total time taken for 2073600 transfers = 2073600*222.5ns = 0.461376 sec > 1/30sec